

Gabriel G. Olafsson

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EDUCATION

Worcester Polytechnic Institute

Worcester, MA

Bachelor of Science in Computer Science, High Distinction

May 2026

EXPERIENCE

Backend & Cloud Infrastructure — IT Lab Systems Intern

Dec 2025 – Jan 2026

New England Biolabs

Ipswich, MA

- Designed a reusable Python-based integration framework connecting core business systems with lab research software, reducing the number of tools scientists needed to manually cross-reference
- Architected Azure cloud infrastructure to support secure, scalable deployment of integration services
- Established standardized testing and deployment practices using GitHub Actions and Pytest, reducing setup time and ensuring consistency across future integrations

Full Stack & DevOps — IT Operations Intern

May – Aug 2025

New England Biolabs

Ipswich, MA

- Deployed real-time network monitoring for 100+ devices using Prometheus and Grafana, secured via an Nginx reverse proxy and Azure AD SSO enforcing role-based view/edit access
- Built a custom React admin interface, backed by a Flask REST API and PostgreSQL database, enabling admins to manage credentials and dynamically define monitored devices
- Implemented containerized architecture with isolated Docker networks to maintain security while operating on a single Virtual Machine

IT Helpdesk Technician Intern

May 2023 – Jan 2025

New England Biolabs

Ipswich, MA

- Built a Python script with pandas and PyAutoGUI to analyze RingCentral phone line usage and auto-draft license-removal outreach emails, cutting license costs by \$15,000+ annually
- Supported 500+ employees with tier 1/2 IT issues, communicating solutions in clear, jargon-free terms

PROJECTS

U.S. Congress Age Data Visualization

Mar – May 2026

- Led a 3-person Agile/Scrum team to deliver an interactive Svelte data visualization on U.S. Congress age trends
- Built a JSON-based ETL pipeline to aggregate, clean, and normalize congressional data from public sources
- Designed the primary visualization's styling and an interactive timeline for exploring age distribution by year

Physical Task Manager For Kanban Board

Jul 2026 – Present

- Built a full-stack task management system with a Rust/Axum backend and Diesel ORM over PostgreSQL, exposing a REST API callable from any device on the network
- Designed a recurrence engine using RRULE to schedule repeating tasks, driving a Raspberry Pi-controlled receipt printer that physically prints tasks on schedule
- Containerized and deployed on Kubernetes for reliable communication across web, database, and printer client

Home Lab Infrastructure

2020 – Present

- Maintain a 3-node Talos Kubernetes cluster hosting a Prometheus/Grafana monitoring stack, and Pi-hole for network-wide DNS filtering, and self-hosted applications
- Configured Nginx for ingress/load balancing and Longhorn for distributed NAS-like persistent storage across nodes

Retro Game Console Inspired Portfolio/Personal Website

Feb – Sep 2026

- Designed a custom retro-console-inspired web page using Astro, Three.js for model animations, and SVG graphics, with fully responsive scaling across desktop and mobile viewports
- Architected a markdown-driven content pipeline where publishing an article only requires adding a markdown file and dropping in images (auto-compressed via ImageMagick on deployment), without having to edit any code

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript, JavaScript, HTML, CSS

Backend: Flask, FastAPI, Axum, PostgreSQL, MongoDB, REST APIs

Infra: Docker, Kubernetes, Azure, Nginx, CI/CD (GitHub Actions), Linux